0:00

hi

0:04

this is tom from zelda finance calm this

0:07

video is designed to cover the different

0:08

cells involved in the immune system so

0:11

these are the cells responsible for

0:13

body's defenses to pathogens and are

0:16

responsible for destroying abnormal

0:18

cells such as virally infected cells or

0:21

cancer cells specifically

0:23

we're going to look at the family tree

0:25

of these cells of this importance

0:27

understand where these cells come from

0:29

how they develop and differentiate and

0:31

how they relate to each other this will

0:34

give you a good overview and reference

0:36

point for later videos where we'll go

0:38

into detail about how these cells

0:39

interact to provide an immune response

0:42

so the development of immune cells

0:45

happens in the bone marrow and it starts

0:48

with a pluripotent hematopoietic stem

0:51

cell so Clarie meaning multiple potent

0:55

meaning potential he motto meaning

0:57

relating to blood and poetic meaning

1:01

creative and then stem cell being an

1:03

undifferentiated cell that has the

1:06

potential to create multiple different

1:08

types of blood cell

1:10

so when this cell is stimulated by

1:13

chemicals called differentiation

1:14

inducers it starts to specialize in to a

1:18

variety of different types the first

1:20

subdivision is to become either a

1:23

myeloid stem cell or a lymphoid stem

1:26

cell now myeloid refers to the bone

1:30

marrow and lympho refers to the

1:33

lymphatic system

1:35

so first let's look at myeloid stem

1:38

cells as well as the cells of the immune

1:41

system myeloid stem cells are also

1:44

responsible for producing platelets and

1:46

red blood cells so they can become

1:49

something called megakaryocytes and mega

1:52

means obviously large and carrier refers

1:56

to the fact that it has a nucleus and

1:58

site meaning cell so as you might

2:01

imagine this is a large cell with a big

2:04

lobulated nucleus and it's responsible

2:07

for producing platelets and platelets

2:10

obviously essential for forming blood

2:11

clots

2:15

so the myeloid stem cell can also become

2:18

reticulocytes reticular

2:20

means net like and sight means cell so

2:24

these are immature red blood cells and

2:26

they've lost their nucleus but they

2:28

still contain remnant RNA material

2:31

called reticulum and this RNA material

2:34

as you can see in the picture is sort of

2:36

net like

2:38

so these reticulocytes are released from

2:41

the bone marrow into the bloodstream

2:43

where they gradually lose their

2:45

reticulum and change shape becoming the

2:49

classical donut like red blood cells the

2:51

red blood cells themselves can survive

2:54

up to three months in the blood and they

2:56

contain a lot of a protein called

2:57

hemoglobin which is the molecule that

3:00

binds to oxygen and allows the red blood

3:02

cell to carry oxygen around the body

3:06

so normally about 1% of the red blood

3:09

cells or reticulocytes and where you see

3:12

a larger proportion of reticulocytes to

3:14

red blood cells you know that the bone

3:16

marrow is having a rapid turnover of red

3:20

blood cells and that'll be because of

3:22

situations like acute blood loss or

3:24

hemolysis where the red blood cells are

3:27

being broken down and this causes an

3:30

increased rate of production and

3:32

therefore a higher proportion of

3:33

reticulocytes to red blood cells

3:38

so alternatively myeloid stem cells can

3:41

become milah blasts and the maiale

3:43

refers to the bone marrow and blast

3:46

refers to immature cells so these

3:48

myeloblast themselves then differentiate

3:51

into other immune cells

3:55

so they can differentiate into monocytes

3:58

and these monocytes then become

4:01

macrophages so mono

4:04

refers to the single nucleus sight

4:07

refers to cell macro refers to the fact

4:10

that it's large and phage means to eat

4:13

so

4:14

monocytes circulate through the

4:16

bloodstream and then when they enter the

4:19

tissues they actually differentiate into

4:21

the macrophages so these are very

4:23

similar the same family of cell

4:27

macrophages can be found in most tissues

4:30

in the body and they simply wait in

4:32

those tissues to encounter a pathogen

4:34

they're the first line of defense for a

4:37

pathogen that has penetrated through the

4:39

barrier defenses of the body now they

4:42

have receptors called toll-like

4:44

receptors and these are essential in

4:47

recognizing molecules that are unique to

4:50

pathogens and these molecules are known

4:52

as pathogen associated molecular

4:54

patterns or PA MPs

4:58

once they recognize a pathogen or

5:00

abnormal cell they can then destroy them

5:03

using a process called phagocytosis

5:04

where

5:06

they essentially absorb or eat the

5:08

pathogen and then digest it from the

5:10

inside

5:13

they're also essential and secreting

5:14

cytokines that initiate the inflammatory

5:17

response when these cytokines and

5:20

inflammation alert the rest of the

5:21

immune system of the infection and then

5:24

they generate the immune response

5:27

so they're the first step in the innate

5:30

immune system it's a response to a new

5:33

pathogen and they alert everybody else

5:37

so myeloblast can also become

5:40

neutrophils neutro refers to neutral and

5:43

fill meaning attraction and this

5:47

therefore refers to the fact that

5:49

neutrophils are generally attracted to

5:51

pathogens which described their role as

5:54

a general attack cell in the innate

5:56

immune system they spend their time

5:58

circulating through the bloodstream and

6:00

then they migrate to areas of

6:01

inflammation in response to signals from

6:04

macrophages and there they destroy the

6:08

pathogens and abnormal cells again by

6:10

phagocytosis

6:14

my Oblast can also become something

6:16

called a sinner Phil's a sinner refers

6:19

to red-colored referring to the fact

6:22

that their cytoplasmic granules take up

6:25

dyes really well and give a red color

6:27

under a microscope the

6:30

sinner Phil's are particularly important

6:32

when the invading pathogens are too

6:33

large to undergo phagocytosis such as in

6:37

parasitic infections so

6:40

they normally hang around in tissues

6:41

until they're activated by cytokines or

6:44

they respond to an invading pathogen

6:48

so they have granules that contain

6:51

pro-inflammatory cytokines

6:54

and chemicals that are directly toxic to

6:56

pathogens and parasites such as major

7:00

basic protein which is an enzyme that

7:02

digests parasites these chemicals can be

7:05

released in a process known as

7:07

degranulation or exocytosis in which the

7:10

granules fuse with the cell membrane and

7:13

then they're released into the

7:14

surrounding space where they can travel

7:16

to cause damage to nearby pathogens

7:22

these cinder fills are also capable of

7:25

phagocytosis but not usually to the

7:27

extent of macrophages or neutrophils

7:31

so the myeloblasts can also become

7:33

Marcel's and basophils and these are two

7:37

very similar types of cell that both

7:39

contain cytoplasmic granules that have

7:42

various cytokines in them and they can

7:44

be released in response to a pathogen

7:46

again by degranulation

7:49

the main difference between these two is

7:51

that Marcel's are fixed in place in the

7:54

tissues whereas basophils circulate

7:57

through the blood

7:59

and that basically concludes all of the

8:01

cells that result from the myeloid stem

8:03

cell and these cells are responsible for

8:06

is called the innate immune system so

8:09

the innate immune system forms the first

8:10

line of defense to pathogens within the

8:12

body and involves the mechanisms for

8:15

general recognition and destruction of

8:17

pathogens

8:21

so next let's look at the cells of the

8:24

specific immune system and these results

8:26

from the lymphoid stem cell so the

8:29

lymphoid stem cell through a series of

8:31

intermediate steps including

8:33

lymphoblasts can become one of three

8:35

types of lymphocyte the first is the B

8:38

lymphocyte or B cells they're

8:41

called B lymphocytes because they mature

8:43

into this type of lymphocyte within the

8:45

bone marrow itself

8:47

our b-lymphocytes have antibodies also

8:50

called immunoglobulins lining their cell

8:53

surface these

8:55

antibodies are matched to specific

8:57

antigens that are displayed on pathogens

9:01

so each lymphocyte has a single specific

9:04

type of antibody that matches a single

9:07

antigen

9:10

there are millions of different possible

9:12

antibodies and therefore millions of

9:15

different specific b-cells so each

9:18

b-cell is only capable of recognizing

9:20

the specific pathogen that displays the

9:23

antigen that matches their antibody

9:27

the role of the B cell is that once it

9:30

is activated by the specific antigen it

9:33

multiplies furiously and becomes either

9:36

plasma cells or memory b-cells plasma

9:40

cells produce fast quantities of

9:42

antibodies against the pathogen and the

9:45

memory b-cells hang around long after

9:48

the pathogen has been cleared so that

9:51

they can respond much quicker when they

9:53

encounter that pathogen again

9:56

the second type of lymphocyte is the T

9:59

lymphocytes or T cells and the code T

10:02

cells because they actually leave the

10:05

bone marrow has PROFIBUS sites and

10:07

traveled to the thymus gland where they

10:10

mature into T cells

10:13

details have t-cell receptors on their

10:16

cell surface that are specific in the

10:18

same way as antibodies on b-cells to a

10:20

specific antigen and so therefore only

10:23

the t-cell specific to the invading

10:26

pathogen will become active in an

10:28

infection

10:30

so t-cells fall into two categories

10:33

based on the type of receptors that they

10:35

display the t-cells that display cd4

10:39

receptors are called cd4 cells and the T

10:43

cells that display the cd8 receptors are

10:47

called cd8 cells and

10:49

importantly cd4 cells respond to MHC

10:54

class 2 receptors on other cells of the

10:57

body and the cd8 cells respond

11:01

to MHC class 1 receptors on other cells

11:04

and this is important to remember and

11:07

will become a lot clearer over time as

11:09

we go through some of the other details

11:11

of the specific immune system so cd4

11:15

cells become T helper cells when they're

11:17

activated many cells in the immune

11:20

system rely on these T helper cells to

11:22

provide signals that then activate them

11:25

including the cd8 cells so

11:29

with the help of T helper cells cd8

11:31

cells become activated when their T cell

11:34

receptors match a pathogen and they

11:37

multiply very quickly and differentiate

11:39

into cytotoxic T cells

11:42

so cytotoxic t-cells can destroy virally

11:46

infected or otherwise infected other

11:48

cells in two ways the first way is to

11:51

attach themselves to the cell and spray

11:54

it with proteins that are destructive to

11:56

that cell's membrane and therefore

11:59

destroy the cell by cell lysis or they

12:04

can activate the fast pathway FAS

12:07

pathway and this pathway is like a

12:09

self-destructive process and it causes

12:12

the cell to destroy itself by apoptosis

12:16

the third type of lymphocyte is called

12:19

natural killer cells

12:22

these cells are activated by cytokines

12:25

from macrophages and by interferons

12:29

and they recognize infected or abnormal

12:31

cells like the cells of tumors without

12:34

requiring the MHC or specific antigen

12:38

receptors

12:40

they can then kill these cells by

12:42

releasing cytokines that induce cell

12:44

death again by apoptosis they

12:48

can also secrete something called

12:49

interferon gamma that further activates

12:52

macrophages so by creating this cycle of

12:55

macrophages releasing sides of kinds

12:57

that then activate the natural killer

13:00

cells who then release the interferon

13:02

gamma that further activates macrophages

13:05

they create a cycle of positive feedback

13:08

that increases the inflammatory response

13:11

after contact with a pathogen

13:15

so that concludes all the cells that

13:17

result from the lymphoid stem cells and

13:19

the cells that are involved in the

13:21

innate immune system and there's one

13:23

final cell type that we need to consider

13:25

which is very important within the

13:28

immune response and that's dendritic

13:30

cells

13:33

there are various types of dendritic

13:35

cell and they all originate from

13:36

different precursor cells however they

13:39

perform a very similar function

13:40

specialist

13:42

dendritic cells that are found in the

13:44

skin are called Langerhans cells

13:47

so dendritic cells are responsible for

13:50

taking up antigens in the infected

13:52

tissue processing them and then

13:54

presenting the antigens on their cell

13:57

membrane for other cells of the specific

14:00

immune system to recognize and respond

14:03

to they have both MHC class 1 and class

14:06

2 molecules on their cell membranes so

14:10

they're able to present antigens to both

14:13

cd4 and cd8 t-cells they act very much

14:18

like postman delivering messages that a

14:20

specific type of pathogen has infected a

14:23

tissue somewhere else in the body they

14:25

pick up these antigens in the infected

14:27

tissue travel through the lymphatic

14:30

system and the blood to the lymphoid

14:32

tissue such as the lymph nodes and

14:34

spleen and there they deliver the

14:36

antigens to the cd4 and cd8 cells and by

14:40

doing this they play an essential role

14:42

in initiating the specific immune

14:45

response so

14:47

that summarizes an overview of the cells

14:49

of the immune system I hope you found

14:52

this video helpful it was created as

14:54

part of a set of videos that covers

14:56

everything you need to know about the

14:58

immune system for medical school and

14:59

practicing very well as a doctor so go

15:02

ahead and check out the other videos in

15:03

this series

15:04

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questions and a blog all of which has

15:18

been purposefully designed and built

15:20

from the ground up to help you prepare

15:22

for your medical exams and for your

15:24

future career as a doctor and I'll see

15:26

you with the next video